

SUPRIM TABLET

DESCRIPTION

Suprim 480 mg Tablet:
Round, white uncoated tablet, flat faces with "HOVID" embossed on one face and scored on the other.

COMPOSITION

Suprim 480 mg Tablet:
Sulfamethoxazole 400 mg and trimethoprim 80 mg/tablet.

PHARMACODYNAMICS

Mode of Action Co-Trimoxazole is an antibacterial drug composed of two active principles, sulfamethoxazole and trimethoprim. Sulfamethoxazole is a competitive inhibitor of dihydropteroate synthetase enzyme. Sulfamethoxazole competitively inhibits the utilisation of para aminobenzoic acid (PABA) in the synthesis of dihydrofolate by the bacterial cell resulting in bacteriostasis. Trimethoprim binds to and reversibly inhibits bacterial dihydrofolate reductase (DHFR) and blocks the production of tetrahydrofolate. Depending on the conditions the effect may be bactericidal. Thus trimethoprim and sulfamethoxazole block two consecutive steps in the biosynthesis of purines and therefore nucleic acids essential to many bacteria. This action produces marked potentiation of activity in vitro between the two agents.

Mechanism of resistance.

In vitro studies have shown that bacterial resistance can develop more slowly with both sulfamethoxazole and trimethoprim in combination that with either sulfamethoxazole or trimethoprim alone.

Resistance to sulfamethoxazole may occur by different mechanisms. Bacterial mutations cause an increase the concentration of PABA and thereby out-compete with sulfamethoxazole resulting in a reduction of the inhibitory effect on dihydropteroate synthetase enzyme. Another resistance mechanism is plasmid-mediated and results from production of an altered dihydropteroate synthetase enzyme, with reduced affinity for sulfamethoxazole compared to the wild-type enzyme. Resistance to trimethoprim occurs through a plasmid-mediated mutation which results in production of an altered dihydrofolate reductase enzyme having a reduced affinity for trimethoprim compared to the wild-type enzyme.

Trimethoprim binds to plasmodial DHFR but less tightly than to bacterial enzyme. Its affinity for mammalian DHFR is some 50,000 times less than for the corresponding bacterial enzyme.

Many common pathogenic bacteria are susceptible in vitro to trimethoprim and sulfamethoxazole at concentrations well below those reached in blood, tissue fluids and urine after the administration of recommended doses. In common with other antibiotics, however, in vitro activity does not necessarily imply that clinical efficacy has been demonstrated and it must be noted that satisfactory susceptibility testing is achieved only with recommended media free from inhibitory substances, especially thymidine and thymine.

Breakpoints

EUCAST

Enterobacteriaceae: Ss 2 R> 4
S. maltophilia: Ss 4 R> 4
Acinetobacter: Ss 2 R> 4
Staphylococcus: Ss 2 R> 4
Enterococcus: Ss 0.032 R> 1
Streptococcus ABCG: Ss 1 R> 2
Streptococcus pneumoniae: Ss 1 R> 2
Haemophilus influenza: Ss 0.5 R> 1
Moraxella catarrhalis: Ss0.5 R>1
Pseudomonas aeruginosa and other non-*enterobacteriaceae:* Ss 2* R> 4*

S = susceptible, R = resistant.

*These are CLSI breakpoints since no EUCAST breakpoints are currently available for these organisms.
Trimethoprim: Sulfamethoxazole in the ratio 1:19. Breakpoints are expressed as trimethoprim concentration.

Antibacterial Spectrum

The prevalence of resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable. This information gives only an approximate guidance on probabilities whether microorganisms will be susceptible to trimethoprim/sulfamethoxazole or not.

Trimethoprim/sulfamethoxazole susceptibility against a number of bacteria are shown in the table below:

Commonly susceptible species:
Gram-positive aerobes: <i>Staphylococcus aureus</i> <i>Staphylococcus saprophyticus</i> <i>Streptococcus pyogenes</i>
Gram-negative aerobes: <i>Enterobacter cloacae</i> <i>Haemophilus influenzae</i> <i>Klebsiella oxytoca</i> <i>Moraxella catarrhalis</i> <i>Salmonella</i> spp. <i>Stenotrophomonas maltophilia</i> <i>Yersinia</i> spp.
Species for which acquired resistance may be a problem:
Gram-positive aerobes: <i>Enterococcus faecalis</i> <i>Enterococcus faecium</i> <i>Nocardia</i> spp. <i>Staphylococcus epidermidis</i> <i>Streptococcus pneumoniae</i>
Gram-negative aerobes: <i>Citrobacter</i> spp. <i>Enterobacter aerogenes</i> <i>Escherichia coli</i> <i>Klebsiella pneumoniae</i> <i>Klebsiella pneumonia</i> <i>Proteus mirabilis</i> <i>Proteus vulgaris</i> <i>Providencia</i> spp.
Inherently resistant organisms:
Gram-negative aerobes: <i>Pseudomonas aeruginosa</i> <i>Shigella</i> spp. <i>Vibrio cholera</i>

PHARMACOKINETICS

After oral administration trimethoprim and sulfamethoxazole are rapidly and nearly completely absorbed. The presence of food does not appear to delay absorption. Peak levels in the blood occur between one and four hours after ingestion and the level attained is dose related. Effective levels persist in the blood for up to 24 hours after a therapeutic dose. Steady state levels in adults are reached after dosing for 2-3 days. Neither component has an appreciable effect on the concentrations achieved in the blood by the other.

Trimethoprim is a weak base with a pKa of 7.4. It is lipophilic. Tissue levels of trimethoprim are generally higher than corresponding plasma levels, the lungs and kidneys showing especially high concentrations. Trimethoprim concentrations exceed those in plasma in the case of bile, prostatic fluid and tissue, saliva, sputum and vaginal secretions. Levels in the aqueous humor, breast milk, cerebrospinal fluid, middle ear fluid, synovial fluid and tissue (intestinal) fluid are adequate for antibacterial activity. Trimethoprim passes into amniotic fluid and foetal tissues reaching concentrations approximating those of maternal serum.

VISUPxx-04 (EXP)

Approximately 50% of trimethoprim in the plasma is protein bound. The half-life in man is in the range 8.6 to 17 hours in the presence of normal renal function. It is increased by a factor of 1.5 to 3.0 when the creatinine clearance is less than 10 ml/minute. There appears to be no significant difference in the elderly compared with young patients.

The principal route of excretion of trimethoprim is renal and approximately 50% of the dose is excreted in the urine within 24 hours as unchanged drug. Several metabolites have been identified in the urine. Urinary concentrations of trimethoprim vary widely.

Sulfamethoxazole is a weak acid with a pKa of 6.0. The concentration of active sulfamethoxazole in a variety of body fluids is of the order of 20 to 50% of the plasma concentration.

Approximately 66% of sulfamethoxazole in the plasma is protein bound. The half-life in man is approximately 9 to 11 hours in the presence of normal renal function.

There is no change in the half-life of active sulfamethoxazole with a reduction in renal function but there is prolongation of the half-life of the major, acetylated metabolite when the creatinine clearance is below 25 ml/minute.

The principal route of excretion of sulfamethoxazole is renal; between 15% and 30% of the dose recovered in the urine is in the active form. In elderly patients there is a reduced renal clearance of sulfamethoxazole.

INDICATIONS

For treatment of:

- Urinary tract infections caused by *E. coli*, *Klebsiella*, *Enterobacter*, *P. mirabilis*, *P. vulgaris* and *Morganella morganii*.
- Acute otitis media in children caused by *H. influenzae* and *Streptococcus pneumoniae*.
- Acute exacerbations of chronic bronchitis in adults caused by *H. influenzae* and *Streptococcus pneumoniae*.
- Enteritis caused by *Shigella flexneri* and *S. sonnei*.
- Pneumonitis caused by *Pneumocystis carinii*.
- gonorrhoea caused by gonorrhoeae
- brucellosis caused by *Brucella*.

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients.
- Co-Trimoxazole should not be given to patients with a history of hypersensitivity to sulphonamides and trimethoprim
- Contraindicated in patients showing marked liver parenchymal damage.
- Contraindicated in severe renal insufficiency where repeated measurements of the plasma concentration cannot be performed.
- Co-Trimoxazole should not be given to premature babies nor to full-term infants during the first 6 weeks of life except for the treatment /prophylaxis of PCP in infants 4 weeks of age or greater.

WARNING AND PRECAUTIONS

- Fatallities, although very rare, have occurred due to severe reactions including fulminant hepatic necrosis, agranulocytosis, aplastic anemia, other blood dyscrasias and hypersensitivity of the respiratory tract.
- Life-threatening cutaneous reactions Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) have been reported with the use of co-trimoxazole.
- Patients should be advised of the signs and symptoms and monitored closely for skin reactions. The highest risk for occurrence of SJS or TEN is within the first weeks of treatment.
- If symptoms or signs of SJS or TEN (e.g. progressive skin rash often with blisters or mucosal lesions) are present, cotrimoxazole treatment should be discontinued.
- The best results in managing SJS and TEN come formerly diagnosis and immediate discontinuation of any suspect drug. Early withdrawal is associated with a better prognosis.
- If the patient has developed SJS or TEN with the use of Co-Trimoxazole, this patient must not be re-started in this patient at any time.

Particular care is always advisable when treating elderly patients because, as a group, they are more susceptible to adverse reactions and more likely to suffer serious effects as a result particularly when complicating conditions exist, e.g. impaired kidney and/or liver function and/or concomitant use of other drugs.

An adequate urinary output should be maintained at all times. Evidence of crystalluria in vivo is rare, although sulphonamide crystals have been noted in cooled urine from treated patients. In patients suffering from malnutrition the risk may be increased.

Regular monthly blood counts are advisable when Co-Trimoxazole is given for long periods, or to folate deficient patients or to the elderly, since there exists a possibility of asymptomatic changes in haematological laboratory indices due to lack of available folate. These changes may be reversed by administration of folic acid (5 to 10 mg/day) without interfering with the antibacterial activity.

In glucose-6-phosphate dehydrogenase (G-6-PD) deficient patients haemolysis may occur.

Co-Trimoxazole should be given with caution to patients with severe allergy or bronchial asthma.

Co-Trimoxazole should not be used in the treatment of streptococcal pharyngitis due to Group A beta-haemolytic streptococci; eradication of these organisms from the oropharynx is less effective than with penicillin. Trimethoprim has been noted to impair phenylalanine metabolism but this is of no significance in phenylketonuric patients on appropriate dietary restriction.

The administration of Co-Trimoxazole to patients known or suspected to be at risk of acute porphyria should be avoided. Both trimethoprim and sulphonamides (although not specifically sulfamethoxazole) have been associated with clinical exacerbation of porphyria.

Close monitoring of serum potassium is warranted in patients at risk of hyperkalaemia and hyponatremia. Except under careful supervision Co-Trimoxazole should not be given to patients with serious haematological disorders (see section 4.8). Co-trimoxazole has been given to patients receiving cytotoxic therapy with little or no additional effect on the bone marrow or peripheral blood.

The combination of antibiotics in Co-Trimoxazole should only be used where, in the judgement of the physician, the benefits of treatment outweigh any possible risks; consideration should be given to the use of a single effective antibacterial agent.

Respiratory toxicity

Very rare, severe cases of respiratory toxicity, sometimes progressing to Acute Respiratory Distress Syndrome (ARDS), have been reported during Co-Trimoxazole treatment. The onset of pulmonary signs such as cough, fever and dyspnoea in association with radiological signs of pulmonary infiltrates, and deterioration in pulmonary function may be preliminary signs of ARDS. In such circumstances, Co-Trimoxazole should be discontinued and appropriate treatment given.

Co-Trimoxazole contains lactose monohydrate. Patients with are hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose galactose malabsorption should not take this medicine

PREGNANCY AND LACTATION

There are not any adequate data from the use of Co-Trimoxazole in pregnant women. Case-control studies have shown that there may be an association between exposure to folate antagonists and birth defects in humans.

Trimethoprim is a folate antagonist and, in animal studies, both agents have been shown to cause foetal abnormalities.



Co-Trimoxazole should not be used in pregnancy, particularly in the first trimester, unless clearly necessary. Folate supplementation should be considered if Co-Trimoxazole is used in pregnancy.

Sulfamethoxazole competes with bilirubin for binding to plasma albumin. As significantly maternally derived drug levels persist for several days in the newborn, there may be a risk of precipitating or exacerbating neonatal hyperbilirubinaemia, with an associated theoretical risk of kernicterus, when co-trimoxazole is administered to the mother near the time of delivery. This theoretical risk is particularly relevant in infants at increased risk of hyperbilirubinaemia, such as those who are preterm and those with glucose-6-phosphate dehydrogenase deficiency.

Breast-feeding

The components of co-trimoxazole (trimethoprim and sulfamethoxazole) are excreted in breast milk. Administration of cotrimoxazole should be avoided in late pregnancy and in lactating mothers where the mother or infant has, or is at particular risk of developing, hyperbilirubinaemia. Additionally, administration of co-trimoxazole should be avoided in infants younger than eight weeks in view of the predisposition of young infants to hyperbilirubinaemia.

DRUG INTERACTIONS

Trimethoprim may interfere with the estimation of serum/plasma creatinine when the alkaline picrate reaction is used. This may result in overestimation of serum/plasma creatinine of the order of 10%. The creatinine clearance is reduced: the renal 3 tubular secretion of creatinine is decreased from 23% to 9% while the glomerular filtration remains unchanged.

In some situations, concomitant treatment with zidovudine may increase the risk of haematological adverse reactions to co-trimoxazole. If concomitant treatment is necessary, consideration should be given to monitoring of haematological parameters.

Reversible deterioration in renal function has been observed in patients treated with Co-Trimoxazole and cyclosporin following renal transplantation.

Concurrent use of rifampicin and Co-Trimoxazole results in a shortening of the plasma half-life of trimethoprim after a period of about one week. This is not thought to be of clinical significance.

When trimethoprim is administered simultaneously with drugs that form cations at physiological pH, and are also partly excreted by active renal secretion (e.g. procainamide, amantadine), there is the possibility of competitive inhibition of this process which may lead to an increase in plasma concentration of one or both of the drugs.

In elderly patients concurrently receiving diuretics, mainly thiazides, there appears to be an increased risk of thrombocytopenia with or without purpura.

Occasional reports suggest that patients receiving pyrimethamine at doses in excess of 25mg weekly may develop megaloblastic anemia should Co-Trimoxazole be prescribed concurrently.

Co-Trimoxazole has been shown to potentiate the anticoagulant activity of warfarin via stereo-selective inhibition of its metabolism. Sulfamethoxazole may displace warfarin from plasma-albumin protein-binding sites in vitro.

Careful control of the anticoagulant therapy during treatment with Co-Trimoxazole is advisable.

Co-Trimoxazole prolongs the half-life of phenytoin and if co-administered could result in excessive phenytoin effect. Close monitoring of the patient's condition and serum phenytoin levels are advisable.

Concomitant use of trimethoprim with digoxin has been shown to increase plasma digoxin levels in a proportion of elderly patients.

Co-Trimoxazole may increase the free plasma levels of methotrexate. Trimethoprim interferes with assays for serum methotrexate when dihydrofolate reductase from Lactobacillus casei is used in the assay. No interference occurs if methotrexate is measured by radioimmune assay.

Administration of trimethoprim/sulfamethoxazole 160mg/800mg (co-trimoxazole) causes a 40% increase in lamivudine exposure because of the trimethoprim component. Lamivudine has no effect on the pharmacokinetics of trimethoprim or sulfamethoxazole.

Interaction with sulphonylurea hypoglycaemic agents is uncommon but potentiation has been reported.

Caution should be exercised in patients taking any other drugs that can cause hyperkalaemia.

If Co-Trimoxazole is considered appropriate therapy in patients receiving other anti-folate drugs such as methotrexate, a folate supplement should be considered.

MAIN SIDE/ ADVERSE EFFECTS

Concurrent therapy with the following drugs is not recommended: Aminobenzoic acid (PABA), pyrimethamine (increase antifolate activity, risk of developing megaloblastic anaemia).

Suprim may enhance the effect of concurrently administered drugs :oral anticoagulant (i.e. warfarin), hypoglycaemic agent (i.e. repaglinide), methotrexate, phenytoin, thiopental (anaesthetic), digoxin, lamivudine, therefore may necessitate dosage adjustment.

Close monitoring are advisable in patient receiving concomitant drugs: zidovudine (increase risk of haematological adverse reactions), Cyclosporin (increase risk of nephrotoxicity), diuretics ,mainly thiazides (increase risk of thrombocytopenia,especially in elderly), patient receiving drugs that can cause hyperkalemia.

As co-trimoxazole contains trimethoprim and a sulphonamide the type and frequency of adverse reactions associated with such compounds are expected to be consistent with extensive historical experience.

Data from large published clinical trials were used to determine the frequency of very common to rare adverse events. Very rare adverse events were primarily determined from post-marketing experience data and therefore refer to reporting rate rather than a "true" frequency. In addition, adverse events may vary in their incidence depending on the indication.

Infections and Infestations

Common: Monilial overgrowth

Blood and lymphatic system disorders

Very rare: Leucopenia, neutropenia, thrombocytopenia, agranulocytosis, megaloblastic anemia, aplastic anemia, haemolytic anemia, methaemoglobinaemia, eosinophilia, purpura, haemolysis in certain susceptible G-6-PD deficient patients. The majority of haematological changes are mild and reversible when treatment is stopped. Most of the changes cause no clinical symptoms although they may become severe in isolated cases, especially in the elderly, in those with hepatic or renal dysfunction or in those with poor folate status. Fatalities have been reported in at-risk patients and these patients should be observed carefully.

Immune system disorders

Very rare: Serum sickness, anaphylaxis, allergic myocarditis, angioedema, drug fever, allergic vasculitis resembling Henoch Schoenlein purpura, periarteritis nodosa, systemic lupus erythematosus.

Metabolism and nutrition disorders

Very common: Hyperkalaemia
Very rare: Hypoglycaemia, hyponatraemia, anorexia
Close supervision is recommended when cotrimoxazole is used in elderly patients or in patients taking high doses of cotrimoxazole as these patients may be more susceptible to hyperkalaemia and hyponatraemia.

Psychiatric disorders

Very rare: Depression, hallucinations

SUPRIM TABLET

Nervous system disorders

Common: headache
Very rare: Aseptic meningitis, convulsions, peripheral neuritis, ataxia, vertigo, tinnitus, dizziness
Aseptic meningitis was rapidly reversible on withdrawal of the drug, but recurred in a number of cases on re-exposure to either co- trimoxazole or to trimethoprim alone.

Respiratory, thoracic and mediastinal disorders

Very rare: Cough, shortness of breath, pulmonary infiltrates
Cough, shortness of breath and pulmonary infiltrates may be early indicators of respiratory hypersensitivity which, while very rare, has been fatal.

Gastrointestinal disorders

Common: Nausea, diarrhea

Uncommon: Vomiting

Very rare: Glossitis, stomatitis, pseudomembranous colitis, pancreatitis

Eye Disorders

Very rare: Uveitis

Hepatobiliary disorders

Very rare: Elevation of serum transaminases, elevation of bilirubin levels, cholestatic jaundice, hepatic necrosis Cholestatic jaundice and hepatic necrosis may be fatal.

Skin and subcutaneous tissue disorders

Common: Skin rashes
Very rare: Photosensitivity, exfoliative dermatitis, fixed drug eruption, erythema multiforme; severe cutaneous adverse reactions (SCARs): Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) have been reported.

Musculoskeletal and connective tissue disorders

Very rare: Arthralgia, myalgia

Renal and urinary disorders

Very rare: Impaired renal function (sometimes reported as renal failure), interstitial nephritis

Fatalities associated with the administration of sulphonamides and trimethoprim, either alone or in combination, have occurred due to severe reactions, including Stevens-Johnson syndrome, toxic epidermal necrolysis and other reactions. The drug should be discontinued at the first appearance of skin rash or any sign of adverse reaction.

Effects associated with Pneumocystis jiroveci (P.carinii) Pneumonitis (PCP) management

Very rare: Severe hypersensitivity reactions, rash, fever, neutropenia, thrombocytopenia, raised liver enzymes, hyperkalaemia, hyponatraemia.

At the high dosages used for PCP management severe hypersensitivity reactions have been reported, necessitating cessation of therapy. If signs of bone marrow depression occur, the patient should be given calcium folinate supplementation (5-10 mg/day). Severe hypersensitivity reactions have been reported in PCP patients on re-exposure to co-trimoxazole, sometimes after a dosage interval of a few days.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

For Oral Use.

There have been no studies to investigate the effect of co-trimoxazole on driving performance or the ability to operate machinery. Further a detrimental effect on such activities cannot be predicted from the pharmacology of the drug.

Nevertheless the clinical status of the patient and the adverse events profile of co-trimoxazole should be borne in mind when considering the patients ability to operate machinery.

OVERDOSE AND TREATMENT

Symptoms

Nausea, vomiting, dizziness and confusion are likely signs/symptoms of overdosage. Bone marrow depression has been reported in acute trimethoprim overdosage.

Management

If vomiting has not occurred, induction of vomiting may be desirable. Gastric lavage may be useful, though absorption from the gastrointestinal tract is normally very rapid and complete within approximately two hours. This may not be the case in gross overdosage. Dependent on the status of renal function administration of fluids is recommended if urine output is low. Both trimethoprim and active sulfamethoxazole are moderately dialysable by haemodialysis. Peritoneal dialysis is not effective.

DOSAGE AND ADMINISTRATION

It may be preferable to take Co-Trimoxazole with some food or drink, a full glass (240ml) of water to minimize the possibility of gastrointestinal disturbances.

Acute infection:

Adult and children over 12 years old, standard dosage: 2 tablets every 12 hours. This dosage approximates to 6mg trimethoprim and 30mg sulfamethoxazole per kilogram body weight per 24 hours.

In acute infections, suprim should be continued until the patient has been symptom free for 2 days, the majority will require treatment for at least 5 days. If clinical improvement is not evident after 7 days therapy, the patient should be re-assessed.

Patients with impaired renal function need a reduced of dosage.

Creatinine clearance ml /min.	Recommended dosage
>30	Usual dosage
15 – 30	Half of the usual dosage
<15	Not recommended

Higher doses of co-trimoxazole of 120mg per kg body-weight (using 20 mg trimethoprim and 100 mg sulfamethoxazole per kg of body weight) daily when given in 2 or 4 divided doses for 14 days are used in the treatment of pneumocystis carinii pneumonia; blood concentrations of sulphamethoxazole should be monitored.

Co-Trimoxazole should not be given to infants below 6 weeks because of the risk of kernicterus from the Sulphonamide component.

The information given here is limited. For further information, consult your doctor or pharmacist.

Storage: Store below 25°C. Protect from light and moisture.

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